UNIOSUNTH Nursing Research App

User guide: how to collect questionnaire answers from nurses and get your study results

Web address: **uniosunth-research.vercel.app**

Study: Nursing Workload and Infection Prevention & Control (IPC) Compliance among nurses at UNIOSUN Teaching Hospital, Osogbo.

1. What the app is

This app is a website you use to collect questionnaire answers from nurses. It saves every answer to one central study database and works out the results for you automatically. It works on a phone, tablet or computer.

The app has two sides:

- **The nurse side** is open to anyone and has no login. Nurses fill in the questionnaire here.
- **The researcher side** is locked with a PIN. Only you can see the answers and the results here. You will be given the PIN separately. Keep it private and do not write it on this guide.

The menu has five items. On a phone, two of them have shorter names:

Menu item	Name on a phone	What it is for
Home	Home	The start page. Shows how many questionnaires have been collected so far.
Assess	Assess	The nurse questionnaire.
Submissions	Entries	A list of every questionnaire received. *Needs the PIN.*
Data	Data	Check records and remove mistakes from the results. *Needs the PIN.*
Analysis	Stats	The study results and downloads. *Needs the PIN.*

2. Installing the app on a phone

Installing puts an icon called **NursingTool** on your home screen. The app then opens full screen like a normal app, loads faster and copes better with a weak network. It is free, and nothing is downloaded from the Play Store or App Store.

Android phones (use Chrome)

1. Open **Chrome** and go to **uniosunth-research.vercel.app**
2. A box saying **"Install App"** should appear at the bottom of the screen. Tap **Install**.
3. Tap **Install** again to confirm.
4. The **NursingTool** icon appears on your home screen.

If the box does not appear (for example because someone tapped “Not now”):

1. Tap the **three dots** at the top right of Chrome.
2. Tap **“Install app”** or **“Add to Home screen”**.
3. Tap **Install**.

iPhone and iPad (use Safari)

1. Open **Safari** and go to **uniosunth-research.vercel.app**. Use Safari to be safe; some other browsers on iPhone do not offer this option.
2. Tap the **Share button** (the square with an arrow pointing up) at the bottom of the screen.
3. Scroll down and tap **“Add to Home Screen”**.
4. Tap **Add** at the top right.
5. The **NursingTool** icon appears on your home screen.

> **Good to know on iPhone**
>
> The iPhone never shows an “Install App” box, so you must use the steps above.
>
> The home-screen icon may show a plain letter or a small picture of the page instead of the app’s logo. This is normal and the app works the same.
>
> If you will collect data where the network is poor, install the app this way. iPhones can clear saved website data that has not been used for a while, which could lose answers that are still waiting to upload. The installed app is protected from this.

3. Collecting answers from nurses

Tap **Start New Assessment** on the Home page, or **Assess** in the menu. There are four steps, and a bar at the top shows how far along the nurse is. The whole questionnaire takes about 10 minutes.

> **Do not refresh or close the page until the answers are submitted**
>
> The answers are only kept on the screen until you press Submit in Step 4. If the page is closed or refreshed before then, the answers are lost and the nurse must start again.

Step 1: Participant information

- The app gives the nurse a random code, such as **NRS-7KQ2MX**. No names are collected, so every nurse stays anonymous. The nurse should write the code down if they may want to withdraw later.
- **Ward:** chosen from a list of 18 wards, grouped into 6 hospital areas (see the table below).
- **Current shift:** Morning, Afternoon or Night.
- **Highest nursing qualification:** RN, BNSc, RN+BNSc, MSc or PhD.
- **Years of experience** and **number of patients this shift**.

Every box must be filled in before the nurse can move on.

Hospital area	Wards in the list
Medical	Male Medical Ward, Female Medical Ward, Renal Unit
Surgical	Male Surgical Ward, Female Surgical Ward, Orthopaedic Ward, Burns Unit
Paediatric & Neonatal	Paediatric Medical Ward 1, Paediatric Medical Ward 2, Paediatric Surgical Ward, Special Care Baby Unit (SCBU)
Critical & Peri-operative	Intensive Care Unit (ICU), Theatre
Obstetrics & Gynaecology	Labour Ward, Post-Natal Ward, Gynaecology Ward
Emergency	Accident & Emergency (A&E), Children Emergency Unit (CEU)

Step 2: Workload questions (about 3 minutes)

- 12 statements about how heavy the nurse's **most recent completed shift** was.
- They cover three areas: physical work and tasks, mental and emotional stress, and paperwork and supplies.
- The nurse answers each one from **1 to 5**: 1 = Not at all, 2 = Slightly, 3 = Moderately, 4 = Very much, 5 = Extremely.

Step 3: Infection prevention (IPC) questions (about 4 minutes)

- 20 statements about how often the nurse follows safety practices: gloves, gowns and masks; needles and sharps; cleaning; and hand washing.
- The nurse answers **Never, Rarely, Sometimes** or **Always**.
- A few statements describe bad habits, for example "I reuse single-use gloves". The app scores these the other way round automatically, so "Never" counts as the good answer. You do not need to do anything.

The app will not move to the next step until every question is answered.

Step 4: Results and submit

The nurse sees two scores out of 100%, each with a label and a short explanation. A **higher workload score** means a heavier workload. A **higher IPC score** means better compliance.

Workload score	Label	IPC compliance score	Label
Under 25%	Low	Under 50%	Poor
25% to 49%	Moderate	50% to 69%	Suboptimal
50% to 74%	High	70% to 89%	Satisfactory
75% and above	Very High	90% and above	Optimal

> **Press the green "Submit to Study Database" button**
>
> Nothing is saved until you press it. Wait until the screen says **"Saved to the study database"** before handing the device to the next nurse.

- **No internet?** The screen says **“Recorded — waiting for a connection”**. The answers are safe on that device and will send by themselves when the internet comes back. Until then, do not clear the browser and do not uninstall the app.
- **Print / PDF** gives a copy of that nurse’s result.
- **New Assessment** starts a fresh questionnaire for the next nurse. Always use this between nurses.

4. The researcher pages (locked with a PIN)

Submissions, **Data** and **Analysis** ask for the PIN. Once you enter it, these pages stay open until you close the browser tab.

- To lock them straight away **on a computer**, click the green **Coordinator** button at the top of the screen.
- **On a phone**, open the menu and tap **Lock Coordinator Mode**.
- Always lock the pages or close the tab when you finish, especially on a shared device.

Submissions (called “Entries” on a phone)

- A list of every questionnaire received. Records removed from the results are still listed, clearly marked as excluded.
- Totals at the top show how many were received, how many are counted in the results, and how many are excluded.
- Search by nurse code, ward or shift.
- **Print list** prints a short list. **Print full** prints every answer. **CSV** downloads everything as a spreadsheet.

Data

- A table of every record with its scores. Tap a record to see its full details or print it.
- **Export CSV** downloads the records as a spreadsheet.
- Tick **Show excluded** to also see records that have been removed from the results.

> **Records can never be deleted**
>
> This protects the study from accidents and tampering. If a record was a test, was entered wrongly, or a nurse withdraws from the study, open it and press **“Exclude from the analysis”**. It stays on the list, marked as excluded, but it is no longer counted in any result. You can bring it back at any time with **“Restore to the analysis”**.

Analysis (called “Stats” on a phone)

This page works out the results for you and updates by itself every time a new questionnaire is submitted. Excluded records are not counted. It has six parts:

1. **Characteristics of the sample:** who took part, by ward, hospital area, shift, qualification, experience and number of patients.
2. **Objective 1: Nursing workload.** Average scores, how many nurses fall in each band, which questions scored highest, and comparisons between hospital areas, wards, shifts and qualifications.
3. **Objective 2: IPC compliance.** The same breakdown for the second questionnaire.

4. **The link between workload and IPC compliance:** do nurses with a heavier workload follow safety practices less? Shown as a chart and a table.
5. **Hypothesis testing:** each of the four hypotheses below is marked **Rejected** or **Retained**, with a plain sentence explaining what that means.
6. **Conclusions:** one conclusion for each research question and the hypothesis, with pie charts and a bar chart.

	Hypothesis (in plain words)
H1	Workload scores and IPC compliance scores are not linked.
H2	A nurse's compliance level does not depend on their workload level.
H3	Workload is the same across the 6 hospital areas.
H4	IPC compliance is the same across the 6 hospital areas.

Retained means the data do not show a link or a difference. **Rejected** means the data do show a real link or difference.

> **The tests need at least 10 nurses**
>
> With fewer than 10 records, the tests show **"Not testable"**. The same goes for the check that the questionnaire gives consistent answers (reliability). The more nurses take part, the more trustworthy the results.

Downloading your results

Press **Download Analysis** at the top of the Analysis page. You get four choices:

Choice	What you get
Analysis report (print / PDF)	The full written report, ready to save as a PDF.
Analysis report (HTML file)	The same report as a file that opens in Microsoft Word.
Statistics (CSV)	Every results table as a spreadsheet.
Raw data (CSV)	One row per nurse with every answer, for Excel, SPSS or R.

5. The six golden rules

1. **Always press Submit**, and wait for the green "Saved" message.
2. **Do not refresh or close the page** in the middle of a questionnaire.
3. **One nurse, one questionnaire.** Press "New Assessment" between nurses.
4. **Keep the PIN private**, and lock or close the tab after using the researcher pages.
5. **Mistakes cannot be deleted.** Exclude them instead.
6. **The tests only work once at least 10 nurses** have taken part.

6. If something goes wrong

What happened	What to do
The page was closed or refreshed halfway through.	The answers were not saved. Start again with New Assessment.
The screen says “Recorded — waiting for a connection”.	The answers are safe on that device. Connect to the internet and they will send by themselves. Do not clear the browser.
The screen says “Submission refused”.	Check the internet connection and tap Try Again. If it keeps happening, contact the person who set up the app.
The screen says “Study database not configured” or “Could not read the database”.	Contact the person who set up the app.
Wrong details were entered, or it was a test entry.	Exclude the record on the Data page. Then ask the nurse to complete a new questionnaire if needed.
A nurse wants to withdraw from the study.	Ask for their nurse code, find the record on the Data page and exclude it.
The “Install App” box does not appear.	Follow the steps in Section 2. On iPhone the box never appears.
The print window does not open.	Allow pop-ups for the website in your browser settings, then try again.
You forgot the PIN.	Ask the person who gave it to you. It cannot be reset from inside the app.